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Enterprise Requirements Document

MoU / CRA / CA Management System and Partnership Performance Dashboard

Directorate of Linkages (RPE)

Prepared for	Directorate of Linkages (RPE), JKUAT
Prepared by	JHUB Africa
Document type	Business Requirements Document (BRD) and System Requirements Specification (SRS)
Version	1.0 - Draft for stakeholder review
Date	10 July 2026
Source basis	Linkages Department internal memo dated 9 July 2026 requesting development of an MoU/CRA/CA management system.

Purpose: define a top-tier, auditable, scalable and user-friendly system for managing the complete lifecycle of institutional partnerships, from request initiation to implementation monitoring, renewal and strategic reporting.

Document Control

Item	Details
Document owner	Directorate of Linkages (RPE)
Technical owner	JHUB Africa
Primary business sponsor	Director, Linkages
Primary implementation partner	Director, JHUB
Review stakeholders	DVC RPE, Linkages Directorate, Legal Office, VC Office, researchers, department/faculty representatives, ICT/system administration, M&E officers.
Approval status	Draft for validation and prioritization.

Version History

Version	Date	Prepared/Updated By	Summary of Change
1.0	10 July 2026	JHUB Africa	Initial requirements document prepared from the Directorate request and expanded into a full enterprise-grade requirements specification.



Table of Contents

The contents below are generated for this requirements pack. Page numbering should be refreshed if the document is edited further.

Section	Page
Executive Summary	4
1. Purpose, Source Basis and Goals	4
2. Business Context and Case for Change	5
3. System Vision and Design Principles	5
4. Scope of Work	6
5. Stakeholders and User Classes	7
6. Target Business Process	7
7. Functional Requirements	9
7.1 Researcher and Initiator Portal	9
7.2 Agreement Type and Template Management	9
7.3 Workflow, Approval and SLA Management	9
7.4 Legal Review and Draft Version Control	9
7.5 Validation, Signing and Activation	9
7.6 Executive KPI Summary Dashboard	9
7.7 Lifecycle Tracker	10
7.8 Monitoring and Evaluation Module	10
7.9 Alerts, Notifications and Escalations	10
7.10 Reports, Analytics and Exports	10
7.11 User, Role and Access Management	10
7.12 Audit, Records and Document Repository	10
7.13 Administration and Configuration	10
7.14 Integrations and APIs	11
8. Non-Functional Requirements	11
9. Data Requirements and Data Model	12
10. Security, Privacy and Compliance Requirements	13
11. Dashboard and Reporting Specification	13
12. Workflow and SLA Rules	14
13. Implementation Approach and Deliverables	15
14. Testing, Quality Assurance and Acceptance	15
15. Risks, Assumptions and Dependencies	16
Appendix A: Detailed Requirement Catalogue	17
Appendix B: Source Requirement Traceability	19
Appendix C: Sample Screen Inventory	19
Appendix D: Glossary	19



Executive Summary

The Directorate of Linkages requires a modern MoU/CRA/CA Management System that moves partnership management from manual tracking into a governed, measurable, auditable and executive-ready digital platform. The requested system must enable researchers and staff to initiate agreement requests, allow Linkages and Legal teams to process and monitor them, and give leadership a real-time view of partnership health, value, risk and performance.

This document expands the Directorate request into a complete requirements specification for JHUB Africa. It defines the business need, expected system capabilities, workflow rules, data model, security controls, dashboards, reporting, implementation approach and acceptance criteria.

The system should be implemented as an enterprise workflow and intelligence platform, not merely a document register. It should combine request intake, template generation, legal review, signing follow-up, active agreement repository, monitoring and evaluation, automated alerts, and executive analytics.

Recommended Target Capabilities

Capability Area	Target Outcome
Researcher self-service	Researchers and initiators submit MoU, CRA and CA requests online and track progress without manual follow-up.
Workflow control	Every request is routed to the correct department/faculty, Linkages, Legal and executive signing stage with SLA monitoring.
Legal and version control	Draft versions, legal reviewers, partner counterparts, comments and review days are tracked in one workspace.
Executive dashboard	Leadership views active partnerships, pipeline volume, dormant/at-risk agreements and value generated.
Lifecycle tracker	All agreements are visible by stage, status color, owner, SLA position and next action.
M&E accountability	Every active agreement has an internal champion, partner liaison, target deliverables, actual outputs and evidence.
Automated alerts	Expiry, dormancy, legal-review delay and pending-action alerts drive proactive management.
Records and audit	Signed documents, approvals, versions and key actions are retained with audit trails and role-based access.

Success Definition

- A single source of truth for MoU, CRA and CA records across the full lifecycle.
- Reduced turnaround time through digital submission, routing, status visibility and escalations.
- Accurate executive KPIs available on demand without manual spreadsheet compilation.
- Measurable partnership implementation through M&E targets, actuals and evidence.
- Improved risk management through expiry warnings, dormancy triggers and stalled-review flags.
- Institutional memory preserved through searchable records, audit trails and controlled archiving.

1. Purpose, Source Basis and Goals

1.1 Purpose of this Document

This requirements document is intended to guide planning, design, procurement, development, testing and acceptance of the Linkages MoU/CRA/CA Management System. It translates the Directorate need into a structured specification that can be used by business owners, JHUB Africa developers, system administrators, testers and approving stakeholders.



1.2 Source Basis

The initial business request was issued by the Director, Linkages to the Director, JHUB and requested assistance in developing an MoU Management Dashboard to enhance turnaround time in processing MoU, CRA and CA matters. The attached requirements identify executive KPIs, lifecycle tracking, M&E accountability, automated alerts, researcher submission, editable templates and researcher progress tracking as the core expected capabilities.

1.3 Strategic Goals

1. Accelerate agreement processing by eliminating opaque, paper-driven or email-only workflow dependencies.
2. Improve accountability by assigning each record to an owner, stage, SLA, internal champion and partner liaison.
3. Strengthen strategic oversight by giving leadership portfolio KPIs across active partnerships, pipeline and risk.
4. Convert signed agreements into measurable outputs through targets, actuals and evidence-based M&E.
5. Reduce institutional risk by ensuring expiry, dormancy and stalled-review matters are surfaced early.
6. Create an auditable institutional repository for current and historical agreements.

1.4 System Name

Recommended working name: Linkages Agreement Lifecycle and Performance Management System (L-ALPMS). Public-facing label: Linkages MoU/CRA/CA Management System.

2. Business Context and Case for Change

Partnership agreements are strategic instruments for research, innovation, entrepreneurship, student training, consultancies, scholarships, grants and institutional collaboration. A manual process makes it difficult to know where a request is stuck, which agreements are active, which are at risk, and what value has been generated from signed partnerships.

2.1 Current Pain Points to be Addressed

Pain Point	Impact	System Response
Limited real-time visibility	Leadership and researchers may not know the current stage or responsible office.	Lifecycle tracker and researcher progress portal.
Manual KPI compilation	Executive reporting may rely on spreadsheets and ad-hoc follow-up.	Automated dashboard, filters and drill-down reports.
Legal review bottlenecks	Drafts can stall without a clear SLA or escalation path.	Legal review workspace, days-in-review counter and Red status after threshold.
Weak post-signature tracking	Signed agreements may not translate into measurable outputs.	M&E module with champion, liaison, targets, actuals and evidence.
Expiry and dormancy risk	Agreements can expire or become inactive without early warning.	Automated expiry warning and dormancy trigger.
Template inconsistency	Different users may use outdated or unapproved samples.	Central editable template library with version control.
Records fragmentation	Documents, approvals and emails may be stored in multiple places.	Central repository and audit trail.

2.2 Business Value

- Faster processing and reduced follow-up burden for researchers and Linkages staff.
- Clear service accountability for departments, Legal, Linkages and executive offices.
- Better renewal planning and fewer missed opportunities due to late expiry awareness.
- Evidence-based reporting on internships, publications, consultancies, grants, scholarships and other outputs.



- Improved partner relationship management through named liaisons and searchable partner histories.

3. System Vision and Design Principles

The system should be designed as an enterprise-grade platform with modular services, secure access, configurable workflows and strong reporting. It should be usable by non-technical users, robust enough for institutional records, and flexible enough to support additional agreement types and future integrations.

Recommended Enterprise Architecture

A secure, modular platform with workflow automation, dashboards, document management and integration readiness.

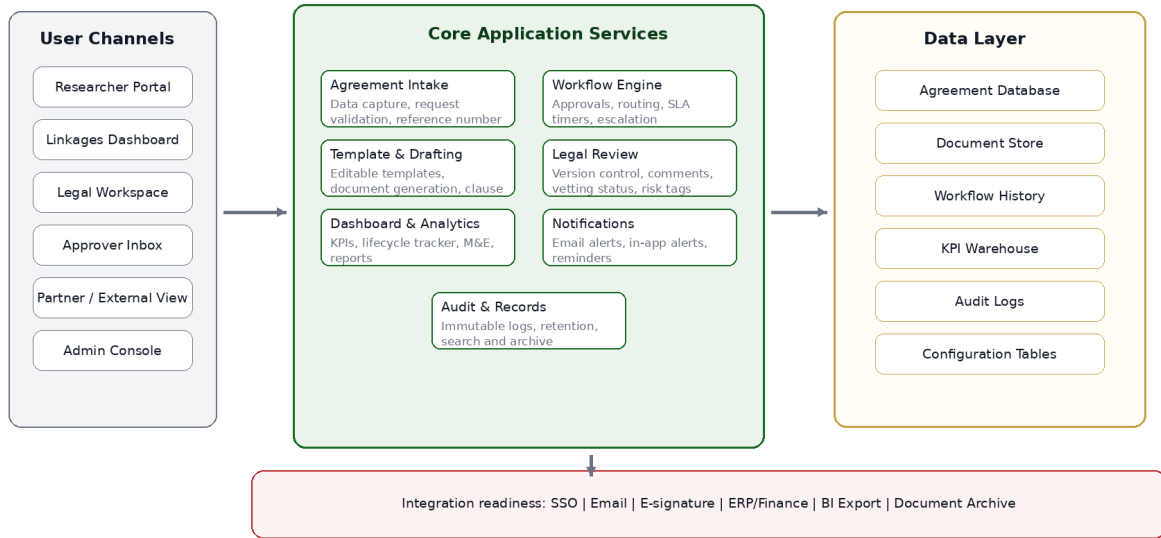


Figure 1: Recommended enterprise architecture for the Linkages Agreement Lifecycle and Performance Management System.

3.1 Design Principles

Principle	Meaning for the System
Single source of truth	One authoritative record per agreement, with complete history and controlled documents.
Workflow-first	Every request must have a stage, owner, next action and SLA state.
Dashboard-driven management	Operational data must feed executive KPIs without manual re-entry.
Configuration over code	Agreement types, templates, SLAs, categories and alerts must be configurable.
Secure by design	Access is controlled by role, department/faculty, document sensitivity and audit policy.
Evidence-based M&E	Partnership performance is measured through targets, actuals and verifiable outputs.
Integration-ready	Architecture should support SSO, email, e-signature, ERP/finance and BI integration where approved.

4. Scope of Work

4.1 In Scope

- Researcher/staff request submission for MoU, CRA and CA matters.
- Agreement template library for editable sample MoU, CRA and CA documents.



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- Partner, sector, department/faculty and strategic alignment data capture.
- Workflow routing through department/faculty approval, Linkages review, Legal review, validation/signing, activation and M&E.
- Executive KPI dashboard, operational lifecycle tracker and M&E dashboard.
- Automated alerts for expiry, dormancy, stalled legal review and pending actions.
- Document repository for drafts, signed documents, supporting evidence and correspondence attachments.
- Reports, exports, audit trail, administrative configuration and role-based access control.
- Migration framework for historical agreement records and documents.

4.2 Out of Scope for Initial Release Unless Approved

- Full contract lifecycle management for procurement contracts unrelated to Linkages partnership agreements.
- Public anonymous partner portal that allows external partners to initiate agreements without JKUAT sponsorship.
- Automated legal approval without human review.
- Direct financial disbursement or grant accounting functions; the system may store value data and integrate later.
- AI-assisted obligation extraction, unless explicitly approved during data governance review.

4.3 Agreement Types

The platform shall support at minimum MoU, CRA and CA. The agreement-type model must be configurable so that Linkages can rename, add or retire types without redevelopment. During discovery, the official expansion and business rules for CRA and CA shall be confirmed and reflected in system labels, templates and reports.

5. Stakeholders and User Classes

Stakeholder / User Class	Primary Interest	Typical System Permissions
Director, Linkages	Portfolio oversight, approvals, escalation, strategic reporting.	Executive dashboard, lifecycle tracker, approvals, reports, configuration oversight.
JHUB Africa	Design, build, test, deploy and support the system.	Technical administration, development, deployment and support access.
DVC RPE	High-level partnership governance and performance visibility.	Executive dashboard, summary reports, escalation views.
Researchers / Initiators	Submit requests and track progress.	Create requests, upload documents, view own records, respond to corrections, update assigned M&E outputs.
Department / Faculty Approvers	Confirm academic fit and departmental support.	Review, approve, return or reject requests within their scope.
Linkages Officers	Process requests, verify completeness, manage partners and coordinate signing.	Full operational workspace for assigned Linkages records.
Legal Reviewers	Draft, vet, comment and approve legal language.	Legal workspace, version control, risk tags, legal decision actions.
VC Office / Executive Signing Desk	Manage final signing and execution follow-up.	Signing tracker, document dispatch, execution date and ceremony details.
M&E Officers / Internal Champions	Track implementation outputs and evidence.	M&E update, deliverable tracking, evidence upload.
System Administrators	Maintain users, roles, templates, categories, SLAs and reference data.	Administration console and technical maintenance functions.
Auditors / Read-only Reviewers	Review compliance, records and performance evidence.	Read-only access to authorized records, reports and audit logs.

5.1 Role-to-Module Access Matrix

Module	Researcher	Dept/Faculty	Linkages	Legal	Executive	Admin
New Request	Create/View own	Review	Create/Edit/Route	View if assigned	View	Configure
Template Library	Use approved	View	Manage with rights	Manage legal clauses	View	Configure
Lifecycle Tracker	Own records	Scope records	Full operational	Legal records	Full read	Full admin



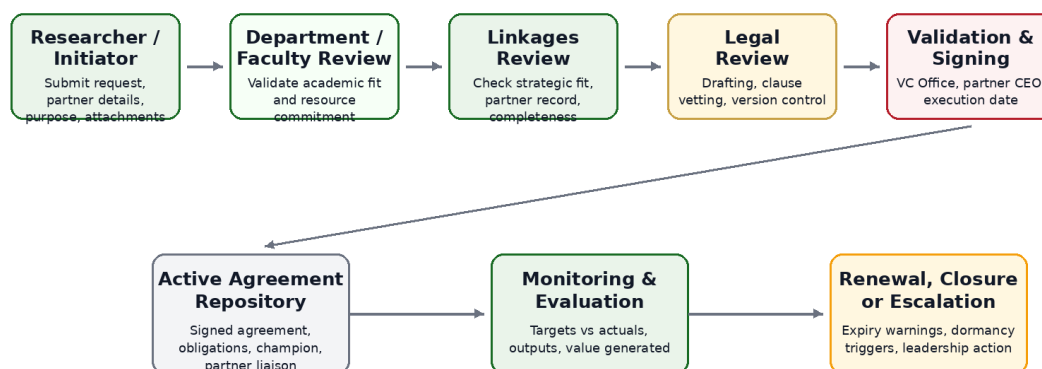
Module	Researcher	Dept/Faculty	Linkages	Legal	Executive	Admin
Legal Workspace	View relevant status	No edit	Coordinate	Full legal edit	View summary	Configure
M&E Module	Assigned updates	Scope view	Full operational	View obligations	Full read	Configure
Reports	Own records	Scope reports	Operational reports	Legal reports	Executive reports	All reports
Audit Logs	No	No	Limited	Limited	Read summary	Full

6. Target Business Process

The target process creates a controlled path from agreement idea to measurable partnership performance. Each stage has a clear responsible office, required data, status indicator and exit criteria.

Target End-to-End Agreement Lifecycle

MoU, CRA and CA requests move through controlled approvals, legal review, signing, activation and measurable delivery.



Mandatory control points: system-generated reference number, SLA timer, audit trail, version history, approval trail and dashboard status indicator.

Figure 2: Target end-to-end agreement lifecycle workflow.

6.1 High-Level Process Steps

Step	Stage	Primary Owner	Exit Criteria
1	Initiation	Researcher / Initiator	Complete request submitted with partner, purpose, agreement type, strategic alignment and attachments.
2	Department/Faculty Approval	Department/faculty approver	Academic fit and departmental support approved or returned/rejected.
3	Linkages Review	Linkages officer / Director	Completeness and strategic alignment validated; record routed to drafting/legal.
4	Drafting and Legal Review	Legal reviewer	Draft vetted, comments resolved, approved version ready for validation and signing.
5	Validation and Signing	Linkages / VC Office / Partner CEO	Dates sent, signature status, ceremony date and fully signed document recorded.
6	Activation	Linkages	Agreement marked Active with effective date, expiry date, champion, partner liaison and deliverables.
7	Monitoring and Evaluation	Internal champion / M&E officer	Targets, actuals and evidence updated on agreed reporting cycle.
8	Renewal, Closure or Archive	Linkages Director / Champion	Renewal decision, closure report or archival action completed.

6.2 Status Color Logic

Indicator	Meaning	Examples
Green	Healthy or approved to proceed.	Approved to proceed; legal vetting active; signing scheduled.



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Indicator	Meaning	Examples
Yellow	Waiting for required action but not yet critical.	Pending departmental approval; awaiting signature.
Orange	Requires proactive management.	Agreement is 6 months from expiry; renewal review due.
Red	Stalled, dormant or high risk.	Legal review exceeds 21 days; no M&E update for 180 days.

7. Functional Requirements

Functional requirements define what the system must do. Priority uses MoSCoW: Must, Should, Could and Won't for current release. Detailed requirement IDs and acceptance criteria are consolidated in Appendix A.

7.1 Researcher and Initiator Portal

- Secure login and personalized dashboard for researchers and initiators.
- Create, save draft, submit and track MoU/CRA/CA requests.
- Guided form with agreement type, partner profile, purpose, strategic alignment, expected outputs and attachments.
- Progress tracker showing current stage, responsible office, action needed, SLA status and comments visible to the initiator.
- Return-for-correction workflow that allows researchers to revise and resubmit without losing history.

7.2 Agreement Type and Template Management

- Editable templates for MoU, CRA and CA, with controlled access and version history.
- Template metadata: agreement type, effective date, owner, approval status and version number.
- Document generation from approved templates using submitted request data.
- Optional clause library for approved standard clauses and special conditions.
- Prevention of uncontrolled template changes by unauthorized users.

7.3 Workflow, Approval and SLA Management

- Configurable workflow stages, role assignments and routing rules.
- Department/faculty approval before Linkages or Legal review where required.
- SLA timers for each stage, including legal review threshold initially set to 21 days.
- Return, reject, escalate, reassign and approve actions with mandatory reason capture where appropriate.
- Complete status history from initiation to archive.

7.4 Legal Review and Draft Version Control

- Legal workspace showing assigned matters, draft versions, days in review and partner legal counterpart.
- Document versioning with version number, uploader, date/time, summary and current approved version.
- Legal comments, risk tags and approval decisions captured as part of the record.
- Stalled-in-legal indicator automatically displayed when review exceeds the configured threshold.
- Read-only preservation of prior versions after a new version is accepted.

7.5 Validation, Signing and Activation

- Capture date sent to Vice Chancellor office, date sent to partner CEO and scheduled signing ceremony date.
- Track signature status: pending VC signature, pending partner signature, fully signed, returned, declined or cancelled.
- Upload final signed PDF or scanned agreement and mark official signed version.
- Activate agreement only after required signature, effective date, expiry date, champion and liaison information are complete.



- Generate activation summary for Linkages and M&E follow-up.

7.6 Executive KPI Summary Dashboard

- KPI card for Total Active Partnerships.
- KPI card for Pipeline Volume covering initiation and drafting phases.
- KPI card for Dormant/At-Risk Counter including zero-activity and expiry-risk agreements.
- KPI card for Total Financial/Resource Value Generated from grants, scholarships, projects or other value records.
- Drill-down from each KPI to filtered records and trend reports.

7.7 Lifecycle Tracker

- Operational table grouped by lifecycle phase and sortable by partner, sector, department/faculty, owner, stage and status.
- Initiation data: partner name, sector, proposing department/faculty, initiator and strategic alignment category.
- Drafting/legal data: draft version, legal reviewer, partner legal counterpart and days in review.
- Validation/signing data: dates sent to VC office and partner CEO, signing ceremony date and signature status.
- Color-coded indicators for healthy, pending, expiry-warning and stalled/dormant matters.

7.8 Monitoring and Evaluation Module

- Dedicated M&E page for every active agreement.
- Accountability fields for internal champion and partner liaison.
- Deliverable checklist with target and actual values for internships, publications, lectures, consultancies, projects and configurable outputs.
- Evidence attachment for each reported output.
- Dormancy logic that turns Red when actual deliverable values have not changed for 180 days.

7.9 Alerts, Notifications and Escalations

- Email and in-app notifications for submissions, approvals, returns, legal assignments, pending signatures, M&E updates and escalations.
- Expiry warning turns Orange six months before agreement expiry.
- Dormancy trigger turns Red after 180 days without actual-output updates.
- Legal stalled alert after 21 days or configured threshold.
- Escalation hierarchy configurable by role, department/faculty and agreement type.

7.10 Reports, Analytics and Exports

- Standard reports: pipeline by stage, active partnerships, expiring agreements, dormant agreements, legal SLA report, M&E output report and value generated report.
- Filter reports by department/faculty, partner sector, agreement type, strategic alignment, date range, status and champion.
- Export to PDF and Excel/CSV with applied filters and generation timestamp.
- Executive trend charts for volume, turnaround time, value and deliverables over time.
- Data quality report for missing champions, expiry dates, partner contacts and deliverables.

7.11 User, Role and Access Management

- Role-based access control for researchers, approvers, Linkages, Legal, executives, M&E, auditors and administrators.
- Organizational scoping by department/faculty and assigned agreement.
- Delegation function for approvers with start/end dates.
- Account provisioning, deactivation and role change audit trail.
- Support for institutional SSO when available; controlled local accounts as fallback if approved.



7.12 Audit, Records and Document Repository

- Central repository for draft documents, final signed agreements, attachments, evidence and exports.
- Full-text and metadata search by partner, agreement type, department, status, champion and dates.
- Immutable audit trail for material actions and record changes.
- Read-only archival of expired, terminated, superseded and closed agreements.
- Records retention and disposal handled according to approved institutional policy.

7.13 Administration and Configuration

- Configurable agreement types, sectors, departments/faculties, strategic alignment categories and partner categories.
- Configurable SLA thresholds and alert rules.
- Configurable notification templates and recipients.
- Template version control and approval workflow.
- System health, backup status and operational settings dashboard for administrators.

7.14 Integrations and APIs

- Integration readiness for institutional identity provider, email service, e-signature platform, ERP/finance data and BI/reporting tools.
- Documented API endpoints or export routines for approved downstream systems.
- Integration error logs and retry handling.
- Manual fallback process when an external service is unavailable.
- Data exchange formats should use standard CSV/Excel/PDF and API JSON where applicable.

8. Non-Functional Requirements

Non-functional requirements define how well the system must operate. They are critical for a top-tier institutional platform because they govern reliability, security, performance, maintainability and user trust.

ID	Category	Requirement
NFR-001	Performance	Core screens shall load within 3 seconds for 95% of normal business-hour interactions under agreed production load.
NFR-002	Dashboard performance	Executive dashboard aggregates shall refresh within 5 seconds for standard filters and within 15 seconds for complex historical filters.
NFR-003	Availability	Target availability shall be at least 99.5% during agreed business hours, excluding planned maintenance.
NFR-004	Scalability	The architecture shall scale to at least 10,000 agreement records, 100,000 documents and 1,000 named users, with final sizing confirmed during discovery.
NFR-005	Security	All user sessions shall use secure authentication, role-based authorization and encrypted transport.
NFR-006	Data protection	The system shall enforce least-privilege access, protect personal data and support compliance with applicable institutional and national data governance requirements.
NFR-007	Auditability	Audit logs shall be tamper-evident and retained according to JKUAT records policy.
NFR-008	Backup and recovery	The system shall support daily backups, tested restoration procedures and a documented recovery time objective.
NFR-009	Usability	Common user tasks shall require minimal steps: draft request creation within one page, status check within one click from dashboard, and M&E update within one form.
NFR-010	Accessibility	User interface shall support keyboard navigation, readable contrast and responsive layouts for desktop and tablet use.
NFR-011	Maintainability	Business rules, templates, SLA thresholds and reference data shall be configurable without code changes.
NFR-012	Interoperability	Reports shall export to Excel/CSV/PDF and APIs shall be designed for future integrations.
NFR-013	Reliability	Submission, approval and upload transactions shall be atomic; partial failures shall not leave records in inconsistent states.



ID	Category	Requirement
NFR-014	Observability	The system shall provide operational logs, error monitoring and administrative health checks.
NFR-015	Browser support	The system shall support current versions of major browsers used by the university, with final compatibility list agreed during discovery.

8.1 Quality Attribute Priorities

Quality Attribute	Priority	Rationale
Security and auditability	Very High	Agreement records, approvals and legal drafts must be protected and traceable.
Usability	Very High	Researchers and officers will adopt the system only if submission and status tracking are simple.
Configurability	High	Agreement types, templates and workflows may evolve.
Dashboard performance	High	Leadership reporting must be fast and trusted.
Integration readiness	Medium-High	Future SSO, e-signature, ERP and BI integrations should not require re-architecture.

9. Data Requirements and Data Model

The system must maintain structured data that supports workflow, dashboards, M&E and records management. The data model should separate partner records, agreement records, workflow history, documents, deliverables, values and audit logs.

9.1 Core Data Entities

Entity	Key Fields	Purpose
Partner	Partner name, sector, type, country, contact persons, legal counterpart, history, status.	Avoid duplicate partner records and enable portfolio analysis by sector.
Agreement Request	Reference number, agreement type, initiator, department/faculty, strategic alignment, purpose, status, dates.	Track intake and processing lifecycle before activation.
Agreement Document	Document ID, version, type, upload date, uploader, file, current/approved flag, confidentiality level.	Manage templates, drafts, signed documents and evidence.
Workflow Event	Record, stage, action, actor, decision, comment, timestamp, SLA clock.	Provide status history, auditability and turnaround analysis.
M&E Deliverable	Agreement, deliverable type, target, actual, period, evidence, last update date, variance.	Measure implementation outputs.
Value Record	Agreement, value type, amount/resource value, currency, source, evidence, reporting period.	Support total financial/resource value generated KPI.
User and Role	User, role, department/faculty, permissions, delegation, status.	Enable secure role-based access.
Audit Log	Actor, action, object, timestamp, old value, new value, source IP/session.	Support compliance, investigation and trust.

9.2 Minimum Agreement Data Dictionary

Field Group	Fields
Identification	System reference number, agreement title, agreement type, record owner, created date, current status.
Partner details	Partner name, sector, partner type, country, partner contact, partner legal counterpart, partner liaison.
JKUAT details	Proposing department/faculty, initiator, internal champion, Linkages officer, legal reviewer, executive signatory.
Strategic classification	Strategic alignment category, priority area, partnership objective, expected outputs.
Lifecycle dates	Submission date, departmental approval date, legal review start/end, date sent to VC Office, date sent to partner CEO, signing date, effective date, expiry date, renewal review date.
Legal and documents	Current draft version, final signed document, template used, legal comments, risk tags, confidentiality level.
M&E	Deliverable types, target, actual, evidence, last updated date, reporting period, implementation notes.
Value	Estimated monetary/resource value, value type, source, currency, evidence and period.
Alerts	Expiry warning flag, dormancy flag, stalled-review flag, escalation history.



10. Security, Privacy and Compliance Requirements

The system will hold institutional records, partner details, legal drafts, approvals and performance evidence. Security must therefore be embedded at application, data, workflow and operational levels.

10.1 Mandatory Controls

Control Area	Requirement
Authentication	Users must authenticate before accessing any protected system page. Institutional SSO is preferred where available.
Authorization	Access must be role-based and scoped by organizational responsibility, assignment and confidentiality.
Least privilege	Default access shall be no access unless explicitly granted through role and scope.
Transport security	All user traffic must be encrypted in transit.
Sensitive documents	Legal drafts, signed agreements and confidential attachments must be access-controlled and logged on download.
Audit trail	Material actions must be recorded and protected against normal-user alteration.
Data retention	Records must be retained, archived and disposed according to approved JKUAT policy.
Backups	Backups must be encrypted or access-controlled and restoration must be tested.
Session controls	Idle sessions should timeout after an agreed period; concurrent-session behavior should be configurable.
Administrative actions	User role changes, configuration changes and template approvals must be logged.

10.2 Compliance Notes

- The system should be aligned with JKUAT ICT, records management, data governance and legal review policies.
- Personal data captured for JKUAT staff, researchers and partner representatives must be limited to necessary business fields.
- External partner access, if enabled, must be strictly scoped to the partner's own records and approved documents.
- Retention schedules and archive rules should be confirmed with the responsible records office during discovery.

11. Dashboard and Reporting Specification

Dashboards must support both executive decision-making and operational daily management. Each metric must have a defined data source, formula and drill-down behavior so that users can trust the numbers.

11.1 Executive KPI Summary

KPI	Definition	Filters / Drill-down
Total Active Partnerships	Count of legally binding operational agreements marked Active and not expired, terminated or archived.	Faculty/department, sector, agreement type, strategic alignment, partner country.
Pipeline Volume	Count of MoU/CRA/CA records in initiation, department approval, Linkages review, drafting/legal or signing stages.	Stage, owner, SLA status, agreement type, department/faculty.
Dormant/At-Risk Counter	Count of active agreements with zero deliverable updates for 180 days, expiring within 90 days, six-month expiry warning or stalled workflow.	Risk type, champion, partner, expiry window, department/faculty.
Total Financial/Resource Value Generated	Sum of approved value records linked to active agreements, including grants, scholarships, resource contributions, joint projects or other defined value types.	Value type, period, currency, agreement type, department/faculty.



11.2 Operational Dashboards and Reports

Dashboard / Report	Users	Purpose
Lifecycle Tracker	Linkages, Legal, leadership	Monitor every agreement by stage, owner, SLA status, risk color and next action.
Legal SLA Report	Legal, Linkages, executives	Show days in review, overdue legal matters and reviewer workload.
Expiry and Renewal Report	Linkages, champions, leadership	Identify agreements expiring in 6 months, 90 days, 30 days or expired.
Dormancy Report	M&E officers, Linkages, champions	Identify active agreements with no recent deliverable progress.
M&E Output Report	M&E, DVC RPE, Linkages Director	Show target versus actual outputs and evidence status.
Value Generated Report	Leadership, Linkages	Summarize monetary and resource value by agreement, partner, department and period.
Data Quality Report	Linkages, administrators	Identify missing expiry dates, missing champions, missing partner contacts and records requiring clean-up.
Audit Report	Administrators, auditors	Review key actions, approvals, edits, downloads and configuration changes.

12. Workflow and SLA Rules

Workflow and SLA rules should be configurable. The initial rules below are recommended based on the Directorate request and should be validated during stakeholder workshops.

12.1 Stage Definitions

Stage	Entry Criteria	Primary Exit Actions	Default Indicator Logic
Initiation	New request created by researcher or Linkages officer.	Submit, save draft, cancel.	Yellow until submitted; Green once complete and routed.
Department/Faculty Approval	Submitted request requires departmental approval.	Approve, return for correction, reject.	Yellow while pending; Green when approved.
Linkages Review	Department/faculty approval complete or Linkages-created request ready for review.	Approve to drafting/legal, return, reject, escalate.	Yellow while pending; Green when approved.
Drafting and Legal Review	Request requires drafting or legal vetting.	Approve draft, return for revisions, request partner input.	Green while active; Red if days in review exceed 21 days.
Validation and Signing	Legally vetted draft approved for signing.	Send to VC Office, send to partner CEO, schedule signing, upload signed copy.	Green if scheduled; Yellow while awaiting signature.
Active Implementation	Final signed agreement uploaded and activation data captured.	Update M&E, renew, close, terminate.	Green if M&E current; Orange on expiry warning; Red on dormancy.
Renewal / Closure / Archive	Agreement nearing expiry, expired, terminated or completed.	Renew, close, archive, supersede.	Orange during renewal review; archived records read-only.

12.2 Recommended Alert Rules

Alert	Trigger	Status	Recipients	Expected Action
Expiry Warning	Current date is six months before expiry date.	Orange	Internal champion, Linkages officer, Linkages Director.	Review renewal, closure or extension.
Expiry Critical	Agreement expires within 90 days.	Red or At-Risk	Champion, Linkages Director, DVC RPE where configured.	Escalate renewal decision.
Dormancy Trigger	No actual deliverable values changed for 180 days.	Red	Champion, partner liaison, M&E officer, Linkages officer.	Update outputs or initiate partner engagement review.
Legal Review Stalled	Drafting/legal review exceeds 21 days.	Red	Legal reviewer, Legal lead, Linkages officer, Linkages Director.	Resolve blocker or escalate.
Department Approval Pending	Approval pending beyond configured department SLA.	Yellow/Red	Department approver, initiator, Linkages officer.	Approve, return or escalate.
Missing Activation Data	Signed agreement uploaded but effective date, expiry date, champion or liaison missing.	Yellow	Linkages officer, initiator.	Complete activation fields.



Alert	Trigger	Status	Recipients	Expected Action
M&E Due	Reporting period reached and actuals not updated.	Yellow	Champion, M&E officer.	Submit target/actual updates and evidence.

13. Implementation Approach and Deliverables

JHUB Africa should implement the system using a phased approach that validates business rules early, delivers high-value workflow functions first and then expands analytics, integrations and advanced automation.

13.1 Recommended Delivery Phases

Phase	Indicative Duration	Key Activities	Main Deliverables
Phase 0: Discovery and Validation	2-3 weeks	Stakeholder workshops, current-process mapping, role validation, data migration assessment, template collection, SLA confirmation.	Approved requirements baseline, process maps, product backlog, data migration plan.
Phase 1: Core Workflow MVP	6-8 weeks	User roles, request submission, department approval, Linkages review, Legal review, basic repository and progress tracker.	Working MVP, workflow engine, request portal, legal workspace, initial templates.
Phase 2: Dashboards and M&E	4-6 weeks	Executive KPIs, lifecycle tracker, M&E module, alert rules, reports and exports.	Executive dashboard, M&E pages, alert engine, standard reports.
Phase 3: Hardening and Migration	3-5 weeks	Security hardening, performance testing, historical data import, UAT fixes, training and documentation.	Migrated data, tested production release, user manuals, admin guide, training materials.
Phase 4: Enhancements	Optional	SSO, e-signature, ERP/finance integration, BI integration, AI-assisted obligation extraction subject to approval.	Integrated and enhanced platform capabilities.

13.2 Expected Project Deliverables

- Stakeholder-approved requirements specification and product backlog.
- Solution architecture, data model, security model and deployment plan.
- Configured workflow stages, roles, SLA rules and alerts.
- Editable MoU, CRA and CA templates configured in the system.
- Researcher portal, Linkages workspace, Legal workspace, executive dashboard and administration console.
- Historical data migration templates and migration report.
- System test report, security test checklist, UAT pack and acceptance sign-off form.
- Training materials, user manuals, administrator guide and support handover pack.

13.3 Training and Change Management

Training should be role-based. Researchers need short submission and tracking guidance; Linkages and Legal users need full workflow and document management training; executives need dashboard and report interpretation; administrators need configuration and operational support training.



14. Testing, Quality Assurance and Acceptance

Acceptance must be based on demonstrable business scenarios, not only screen availability. Each requirement should have test cases showing that the workflow, data, dashboards, alerts and audit controls operate correctly.

14.1 Test Levels

Test Level	Scope
Unit Testing	Developer testing of individual functions, validation rules, services and components.
System Integration Testing	End-to-end testing of workflow, documents, notifications, dashboards, exports and access control.
Security Testing	Authentication, role-based access, data leakage checks, audit logging and administrative controls.
Performance Testing	Page load, dashboard refresh, export processing and concurrent-user scenarios.
Data Migration Testing	Historical records imported accurately, incomplete records flagged and duplicates identified.
User Acceptance Testing	Business users execute real scenarios and confirm readiness for rollout.

14.2 Acceptance Criteria

Area	Acceptance Criteria
Workflow	At least one MoU, one CRA and one CA request can move from initiation through activation using configured approval steps.
Researcher portal	Researcher can submit a request, upload attachments, receive reference number and view progress.
Templates	Approved sample MoU, CRA and CA templates are available and can generate drafts.
Legal review	Draft version, reviewer, counterpart and days-in-review are visible; stalled review turns Red after threshold.
Executive dashboard	Active partnerships, pipeline volume, dormant/at-risk counter and value generated are accurate against test data.
Lifecycle tracker	Records appear in the correct stage with correct color indicators and next-action owner.
M&E	Active agreement has champion, partner liaison, target/actual deliverables and evidence upload.
Alerts	Expiry warning at six months and dormancy after 180 days trigger expected statuses and notifications.
Security	Users cannot view or act on records outside their role and scope.
Audit	Critical actions are logged with actor, date/time and record reference.
Exports	Authorized users can export filtered reports to PDF and Excel/CSV.

14.3 Definition of Done

- All Must requirements have passed test cases or have approved exceptions.
- Critical and high defects are resolved; medium defects have accepted workarounds or delivery dates.
- Role-based access has been tested with representative users.
- Dashboard KPIs reconcile with sample and migrated data.
- User training has been completed for priority roles.
- Production backup, restore, monitoring and support procedures are documented.

15. Risks, Assumptions and Dependencies

15.1 Key Risks and Mitigations

Risk	Potential Impact	Mitigation
Unclear agreement-type definitions for CRA and CA	Incorrect labels, templates or routing.	Confirm official definitions and workflows during discovery; keep agreement types configurable.



Risk	Potential Impact	Mitigation
Incomplete historical data	Dashboard and M&E reports may be inaccurate at launch.	Use migration templates, data quality reports and phased data clean-up.
Resistance to new workflow	Users may continue using email/manual tracking.	Role-based training, leadership sponsorship and mandatory system reference number for processing.
Legal review bottlenecks persist	System surfaces delays but cannot solve resourcing constraints alone.	Use SLA reporting for management action and workload balancing.
Poor M&E update discipline	Dormancy reports may increase without actual implementation action.	Assign champions, automate reminders and include M&E in performance review cycles.
Integration delays	SSO, e-signature or ERP integration may postpone advanced capabilities.	Deliver core workflow first with manual fallback and integration-ready architecture.
Template governance gaps	Unapproved clauses or outdated templates may be used.	Establish template ownership, approval workflow and version control.

15.2 Assumptions

- JKUAT will identify official business owners for workflow stages, templates, alerts and data ownership.
- The Directorate of Linkages will provide current MoU, CRA and CA samples and any approved clause language.
- Legal Office will confirm vetting workflow, sign-off rules and legal status labels.
- ICT or the designated hosting team will confirm deployment environment, domain, backup requirements and identity integration options.
- Historical records can be provided in a structured format or through an agreed data clean-up exercise.
- Final user numbers, concurrency estimates and storage volumes will be confirmed during discovery.

15.3 Dependencies

- Availability of stakeholder representatives for requirements validation and UAT.
- Approval of workflow rules by Linkages, Legal and executive offices.
- Availability of institutional branding, templates and records retention guidance.
- Hosting, security, email and identity-management decisions.
- Timely provision of historical data and signed agreement documents for migration.

Appendix A: Detailed Requirement Catalogue

This catalogue consolidates detailed functional requirements for backlog grooming, estimation, development and testing. Requirements marked Must should be treated as core release acceptance items unless stakeholders formally reprioritize them.

ID	Priority	Requirement	Acceptance Criteria
FR-001	Must	The system shall allow authorized researchers and staff to submit MoU, CRA and CA requests through a secure web portal.	A user can create, save draft, submit and receive a unique system reference number for each request.
FR-002	Must	The request form shall capture partner name, sector, country, contact details, agreement type, purpose, expected outcomes, proposed department/faculty and strategic alignment category.	All mandatory fields are validated before submission and appear on the lifecycle tracker.
FR-003	Must	The system shall support upload of concept notes, draft agreements, partner profiles, correspondence and supporting documents.	Files are stored against the request record with uploader, date, file type, version and access permissions.
FR-004	Must	The system shall provide editable institutional templates for MoU, CRA and CA agreements.	Authorized users can generate a draft from a template; template edits are versioned and permission-controlled.
FR-005	Should	The system shall provide a clause library for standard institutional clauses, renewal provisions, confidentiality clauses and obligation clauses.	Legal reviewers can insert approved clauses into drafts and trace the clause version used.
FR-006	Must	The system shall route requests to department/faculty approvers for preliminary approval.	The request cannot proceed to Linkages review until departmental decision is recorded.
FR-007	Must	The system shall route eligible requests to the Directorate of Linkages for completeness and strategic-fit review.	Linkages officers can approve, return for corrections or reject with reasons.
FR-008	Must	The system shall route agreement drafts to the Legal Office for review and vetting.	Legal reviewer assignment, review start date and review outcome are captured.
FR-009	Must	The system shall track current draft version, JKUAT Legal Team reviewer and partner legal counterpart contact.	The lifecycle tracker displays these fields during drafting and legal review.



ID	Priority	Requirement	Acceptance Criteria
FR-010	Must	The system shall measure days spent in legal review and highlight stalled matters older than the configured SLA threshold.	Matters exceeding 21 calendar days in legal review display Red status unless the SLA threshold is changed by an administrator.
FR-011	Must	The system shall track dates sent to the Vice Chancellor office and to the partner CEO for signature.	Dates are mandatory before a record can move to signing follow-up.
FR-012	Should	The system shall support scheduling and tracking of signing ceremonies where applicable.	Signing ceremony date, venue, participants and status can be recorded.
FR-013	Must	The system shall convert fully signed agreements into active partnership records.	An agreement cannot be marked Active unless final signed document, effective date and expiry date are captured or exemption is approved.
FR-014	Must	The system shall maintain an executive dashboard showing total active partnerships, pipeline volume, dormant or at-risk count and estimated value generated.	Dashboard figures reconcile with underlying agreement records and are filterable by faculty, sector, agreement type and period.
FR-015	Must	The system shall maintain an operational lifecycle tracker covering initiation, drafting/legal, validation/signing, active implementation, renewal and closure.	Each agreement has one current lifecycle status and a complete history of status transitions.
FR-016	Must	The lifecycle tracker shall display status indicators Green, Yellow, Orange and Red based on configured rules.	Color rules are applied automatically and can be audited by an administrator.
FR-017	Must	The system shall maintain an M&E sub-page for every active MoU/CRA/CA.	Each active agreement has accountability metrics, deliverable checklist and output evidence.
FR-018	Must	The M&E module shall capture internal champion and partner liaison for each active agreement.	Both contacts are displayed on the M&E page and included in alerts.
FR-019	Must	The M&E module shall track target versus actual outputs for internships, joint research publications, guest lectures, consultancies, projects and configurable deliverable types.	Users can set annual targets, enter actuals, attach evidence and view variance.
FR-020	Must	The system shall issue an expiry warning when an agreement is six months away from expiry.	The agreement changes to Orange status and renewal review notifications are sent to responsible parties.
FR-021	Must	The system shall issue a dormancy trigger where actual deliverable fields have not changed for 180 days.	The agreement changes to Red status and appears on the dormant/at-risk counter.
FR-022	Must	The system shall include a researcher-facing progress tracker.	Researchers can see current stage, pending office, action required, SLA status and comments relevant to them.
FR-023	Should	The system shall support internal comments, decision notes and return-for-correction loops.	Comments are time-stamped, attributed to a user and visible according to role permissions.
FR-024	Must	The system shall maintain an immutable audit trail for create, edit, submit, approve, reject, upload, download, status change and delete/archive actions.	Audit logs show actor, date/time, record, action and old/new values where applicable.
FR-025	Must	The system shall support role-based access control across all modules.	Users can only view and perform actions allowed by their role and organizational scope.
FR-026	Should	The system shall support delegated acting roles for approvers who are away or unavailable.	Delegations have start and end dates and appear in the audit trail.
FR-027	Must	The system shall provide searchable document repository functions.	Users can search by partner, agreement type, department, status, champion, date, keyword and tags.
FR-028	Should	The system shall generate printable agreement summaries and dashboard reports in PDF and Excel formats.	Authorized users can export filtered views with date, filters and export user indicated.
FR-029	Should	The system shall provide configurable email and in-app notifications.	Notification templates, recipients and triggers can be configured by administrators.
FR-030	Could	The system may support SMS alerts for urgent escalations where institutionally approved.	SMS can be enabled or disabled globally and by event type.
FR-031	Should	The system shall support integration with institutional identity management or single sign-on.	Users can authenticate through the agreed identity provider or controlled local accounts.
FR-032	Should	The system shall support e-signature integration where approved by JKUAT governance.	Signing status can be synchronized or manually recorded depending on integration readiness.
FR-033	Must	The system shall allow administrators to configure faculties, departments, sectors, strategic alignment categories, agreement types, SLA thresholds and alert rules.	Configuration changes are audit logged and take effect without code deployment.
FR-034	Must	The system shall prevent duplicate active partner records unless an authorized user confirms a legitimate separate relationship.	Duplicate warning appears based on partner name and configurable matching rules.
FR-035	Should	The system shall support batch import of historical agreements for migration.	Migration template validates required fields and flags incomplete legacy records.
FR-036	Should	The system shall support data quality dashboards for missing expiry dates, missing champions, missing partner contacts and inactive M&E updates.	Data quality issues are listed with responsible owner and correction status.
FR-037	Must	The system shall maintain agreement financial/resource value records.	Value source, value type, currency, estimated amount, evidence and reporting period can be captured.
FR-038	Could	The system may provide AI-assisted summary and obligation extraction subject to data governance approval.	AI-generated fields remain editable, labelled as generated and cannot overwrite official records without human approval.
FR-039	Must	The system shall archive expired, terminated or superseded agreements while preserving searchability and records retention.	Archived records remain read-only except to authorized records administrators.
FR-040	Should	The system shall support dashboard drill-down from KPI cards to the underlying agreement records.	Clicking a KPI card opens a filtered list of contributing records.



Appendix B: Source Requirement Traceability

The table below shows how the Directorate request has been translated into the expanded requirements in this document.

Source capability / request	Requirement coverage in this document
Develop an MoU Management Dashboard to enhance turnaround time	Workflow automation, SLA tracking, dashboard KPIs, alerts, researcher progress tracking and escalation rules.
Executive KPI Summary: active partnerships, pipeline volume, dormant/at-risk counter, total value generated	Functional requirements FR-014, FR-037 and dashboard specification Section 11.
Lifecycle Tracker: initiation, drafting/legal, validation/signing data fields and indicators	Functional requirements FR-015 and Section 12 workflow/status definitions.
Legal review data: draft version, legal reviewer, partner legal counterpart, days in review	Functional requirements FR-008 to FR-010 and data dictionary Section 9.
Validation/signing data: dates sent to VC Office and Partner CEO, signing ceremony date	Functional requirements FR-011 to FR-013.
M&E: internal champion, partner liaison, deliverable checklist	Functional requirements FR-017 to FR-019 and M&E dashboard definitions.
Automated expiry warning 6 months before expiry	Functional requirement FR-020 and alert rules in Section 12.
Dormancy trigger if actual deliverables do not change in 180 days	Functional requirement FR-021 and alert rules in Section 12.
Researchers to submit MoU, CRA, CA requests through the system	Functional requirements FR-001 to FR-003 and Section 7.1.
Embed editable sample MoU, CRA, CA	Functional requirements FR-004 and FR-005.
Researchers to track progress from their interface	Functional requirement FR-022.

Appendix C: Sample Screen Inventory

The following screen inventory is recommended for the first full release. Screen names may be adjusted during UX design while retaining the same functional coverage.

Screen / Page	Primary Users	Core Purpose
Login / SSO Landing Page	All users	Secure access and routing to the correct dashboard.
Researcher Dashboard	Researchers, initiators	Submit requests, view drafts, track progress and respond to corrections.
New Agreement Request Form	Researchers, Linkages officers	Capture agreement type, partner details, strategic alignment and expected deliverables.
Department Approval Inbox	Department/faculty approvers	Review, approve, return or reject requests.
Linkages Review Workspace	Linkages officers and Director	Completeness checks, partner verification, strategic-fit review and routing.
Legal Review Workspace	Legal reviewers	Draft management, review comments, version control and legal SLA tracking.
VC / Executive Signing Tracker	VC Office, Linkages Director	Track documents sent for signature, ceremony dates and final execution.
Executive KPI Dashboard	DVC RPE, Linkages Director, management	Show partnership portfolio health and strategic value.
Lifecycle Tracker	Linkages, Legal, leadership	Operational view of all agreements by stage, owner, SLA and status indicator.
Agreement Detail Page	Authorized users	Single record view for agreement data, documents, status, comments, and history.
M&E Update Page	Internal champions, M&E officers	Update targets, actual outputs, evidence and partner engagement activity.
Reports and Exports Page	Management, auditors, officers	Generate filtered reports and export data.
Administration Console	System administrators	Configure users, roles, templates, SLA rules and reference data.



Appendix D: Glossary

Term	Definition
MoU	Memorandum of Understanding. A formal agreement describing mutual intent, areas of cooperation and obligations.
CRA	Collaborative/Cooperative Research Agreement, as defined by JKUAT policy. The system shall keep the acronym configurable to match official wording.
CA	Agreement type identified by the Directorate. The full institutional expansion shall be confirmed during discovery and configured in the system.
Pipeline	Agreement requests that are not yet active, including initiation, review, drafting, legal and signing stages.
Active Partnership	A legally binding and operational agreement that has been signed, activated and has not expired, terminated or been archived.
Dormant	An active agreement with no recorded deliverable progress for the configured dormancy period, initially 180 days.
At-Risk	An agreement that is expiring soon, stalled in an approval process, dormant or otherwise flagged by system rules.
Internal Champion	JKUAT lecturer, researcher or officer responsible for day-to-day implementation of an active agreement.
Partner Liaison	Named contact person from the external partner institution.
SLA	Service Level Agreement threshold used to monitor process turnaround and trigger escalations.

Closing Note

This document is intended to be validated by the Directorate of Linkages and relevant executive stakeholders before build commencement.

